Ex.no: 2

Date: 16/7/26

STUDY OF DIFFERENT TYPES OF NETWORK CABLES AND CRIMPING OF CABLES WITH RJ45 CONNECTOR.

Aim:

To study the different types of network cables.

a) Understanding different types of network cables. The different types of cables used in networking are:

1) Unshield twisted Pair (UTP) cable
2) Shielded twisted Pair (STP) cable
3) Coaxial cable.
4) Fibre optic cable.

Cable Type	category	Maximum data transmission	Advantages/ Disadvantages	Applications
UTP	category 3 category 5 category 5e	10 bps upto 100 Mbps 1 Gbps	Advantages: → cheaper → Easy to install Disadvantages: → more prone to EMI & noise.	10 Base-T Ethernet Fast Ethernet
STP	category 6, 6a.	10 Gbps	Advantages: → Shielded → Faster than UTP → less to noise.	Gigabit-Ethernet
coaxial cable	RG1-6 RG1-59 QG1-11	10-100 Mbps	Advantage: → High band width → versatile. Disadvantage: → limited distance. → cost. → size is bulky.	Speed of signal is 500m, Television network
Fibre optic cable	single mode, multimode	100 Gbps	Advantage: → high speed. → high band width. → high security Disadvantage: → Expensive. → Require skilled Installers	maximum distance of fibre optics cable is 100 meter.

Result:

Thus, the different types of network cables were studied and verified.